# Tamsworth Interconnect Co.: Impedance-Controlled Stackup Report

## 1. Document

| Field | Value |
|---|---|
| Fabricator | Tamsworth Interconnect Co. |
| Quote number | TIC-Q-4471 |
| Revision | C |
| Issue date | 2026-10-06 |
| Customer part number | RFX-8L-STK |
| Board size | 50.0 x 40.0 mm, 2.0 mm corner radius |
| Layer count | 8 |
| Construction | 1 + 6 + 1 with a second build-up level on the top side. Stacked laser microvia L1 to L3 |
| Overall thickness | 81.00 mil (including soldermask both sides) |
| Finished thickness | 80.00 mil (copper and dielectric, excluding soldermask) |
| Thickness tolerance | +/-10 % |
| Surface finish | ENIG |
| Soldermask | LPI green, both sides |
| Copper plating class | IPC-6012 Class 2 |
| Impedance tolerance | +/-10 %, all controlled structures |
| Impedance model | TIC 2D field solver |
| Impedance test frequency | 5.0 GHz |
| Dk/Df reference frequency | 2.0 GHz |
| Units | Dimensions are in mil. Where a mm value is also given it is for reference only; the mil value controls. |

## 2. Revision history

| Rev | Date | Change |
|---|---|---|
| A | 2026-08-11 | Initial issue. |
| B | 2026-09-15 | L4-L5 prepreg changed from 3.5 mil to 4.0 mil; the L4 stripline widths re-solved. |
| C | 2026-10-06 | UV-1 capture pad enlarged from 9.0 mil to 10.0 mil for registration margin. No other change. |

## 3. Materials

| ID | Manufacturer | Product | Type | Glass | Resin % | Dk | Df | Tg C |
|---|---|---|---|---|---|---|---|---|
| D-SM | Corvel Coatings | CV-12 | Soldermask | | | 3.60 | 0.0200 | |
| D-BU | Ashgrove Laminates | AL-330 | Prepreg | 1067 | 70 | 3.30 | 0.0025 | 200 |
| D-CORE | Ashgrove Laminates | AL-360 | Laminate core | 1080 | 60 | 3.60 | 0.0080 | 200 |
| D-PP | Ashgrove Laminates | AL-355 | Prepreg | 2113 | 62 | 3.55 | 0.0070 | 200 |
| D-CORE-T | Ashgrove Laminates | AL-370 | Laminate core | 7628 | 45 | 3.70 | 0.0090 | 200 |

Dk and Df at 2.0 GHz.

| ID | Manufacturer | Product | Foil | Weight | Thickness | Rz matte um | Rz drum um |
|---|---|---|---|---|---|---|---|
| CU-OUT | Brightwell Foil | BF-V2 | VLP | 0.5 oz | 1.50 mil finished (base foil plus plating) | 2.5 | 0.8 |
| CU-IN | Brightwell Foil | BF-R1 | RTF | 1 oz | | 4.0 | 2.0 |

## 4. Stackup

| Seq | Type | Layer | Function | Material | Copper | mil | mm | Dk | Df |
|---|---|---|---|---|---|---|---|---|---|
| 1 | Soldermask | | Soldermask top | D-SM | | 0.50 | 0.0127 | 3.60 | 0.0200 |
| 2 | Copper | L1 | Signal - surface microstrip referenced to L3 | CU-OUT | 0.5 oz + plating | 1.50 | 0.0381 | | |
| 3 | Dielectric | | Build-up 1 | D-BU | | 3.00 | 0.0762 | 3.30 | 0.0025 |
| 4 | Copper | L2 | Signal - low speed; voided under L1 controlled routes | CU-IN | 1 oz | | | | |
| 5 | Dielectric | | Build-up 2 | D-BU | | 3.00 | 0.0762 | 3.30 | 0.0025 |
| 6 | Copper | L3 | Plane - GND | CU-IN | 1 oz | | | | |
| 7 | Dielectric | | Core | D-CORE | | 5.00 | 0.1270 | 3.60 | 0.0080 |
| 8 | Copper | L4 | Signal - stripline | CU-IN | 1 oz | | | | |
| 9 | Dielectric | | Prepreg | D-PP | | 4.00 | 0.1000 | 3.55 | 0.0070 |
| 10 | Copper | L5 | Plane - GND | CU-IN | 1 oz | | | | |
| 11 | Dielectric | | Thick core | D-CORE-T | | | | 3.70 | 0.0090 |
| 12 | Copper | L6 | Plane - PWR | CU-IN | 1 oz | | | | |
| 13 | Dielectric | | Core | D-CORE | | 5.00 | 0.1270 | 3.60 | 0.0080 |
| 14 | Copper | L7 | Plane - GND | CU-IN | 1 oz | | | | |
| 15 | Dielectric | | Prepreg | D-PP | | 4.00 | 0.1016 | 3.55 | 0.0070 |
| 16 | Copper | L8 | Signal - surface microstrip | CU-OUT | 0.5 oz + plating | 1.50 | 0.0381 | | |
| 17 | Soldermask | | Soldermask bottom | D-SM | | 0.50 | 0.0127 | 3.60 | 0.0200 |

Layers numbered from the top: L1 is top copper, L8 is bottom copper. L2 is voided under every L1 controlled route.

## 5. Drill and via table

| Via | Structure | Drill | Span | Hole mil (mm) | Pad mil (mm) | Depth mil (mm) | Aspect ratio | Ring mil | Fill | Capped | Tented | Via in pad |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| UV-1 | Microvia | Laser | L1-L2 | 4.00 (0.1016) | 10.00 (0.2540) | 3.00 (0.0762) | 0.75:1 * | 2.50 | Copper | Yes | Both | Yes |
| SV-1 | Stacked microvia, 2 levels | Laser | L1-L3 | 6.00 (0.1524) | 12.00 (0.3048) | 7.38 (0.1874) | 1.23:1 * | 3.00 | Copper | Yes | Both | Yes |
| BD-1 | Backdrilled through | Mechanical | L1-L8, backdrilled from L8 keeping L1-L4 | 10.00 (0.2540) | 20.00 (0.5080) | 80.00 (2.0320) | 8.00:1 * | 5.00 | Resin | Yes | Both | No |
| TH-1 | Through | Mechanical | L1-L8 | 8.00 (0.2032) | 16.00 (0.4064) | 80.00 (2.0320) | 10.00:1 * | 4.00 | None | No | Both | No |

* Aspect ratio = depth / finished hole. Mechanical depth is the full drilled depth, copper at both ends included. Laser depth runs from the bottom of the entry copper to the top of the target copper (end foils excluded).

## 5a. Backdrill

| Via | Side | Keeps | Drill mil | Start pad mil | Mask opening mil | Copper clearance mil |
|---|---|---|---|---|---|---|
| BD-1 | From L8 | Keep L1-L4 | 16.00 | 24.00 | 26.00 | 28.00 |

## 5b. Non-functional pad removal

| Via | Layers | Reduced pad mil (mm) | Functional pad mil (mm) |
|---|---|---|---|
| TH-1 | L2, L3, L4, L5, L6, L7 | 11.00 (0.2794) | 16.00 (0.4064) on every other layer |

## 6. Impedance

| Structure | Layer | Geometry | Target ohm | Width mil (mm) | Gap mil (mm) | Clearance mil (mm) | Reference | Modelled ohm | Eps eff | Delay ps/mm | Loss dB/mm |
|---|---|---|---|---|---|---|---|---|---|---|---|
| SE-50 | L1 | Coated microstrip over L3 (L2 voided) | 50 +/-10 % | 11.00 (0.2794) | | 22.00 (0.5588) | L3 | 50.4 | 2.7000 | 5.4810 | 0.00600 |
| SE-50 | L8 | Coated microstrip | 50 +/-10 % | 7.50 (0.1905) | | 15.00 (0.3810) | L7 | 49.7 | 2.6600 | 5.4403 | 0.00680 |
| SE-50 | L4 | Asymmetric stripline | 50 +/-10 % | 4.50 (0.1143) | | 9.00 (0.2286) | L3+L5 | 50.2 | 3.5800 | 6.3113 | 0.00950 |
| DIFF-100 | L4 | Asymmetric stripline edge-coupled | 100 +/-10 % | 4.00 (0.1016) | 6.00 (0.1524) | 12.00 (0.3048) | L3+L5 | 99.1 | 3.5800 | 6.3113 | 0.01050 |
| DIFF-100 | L8 | Coated microstrip edge-coupled | 100 +/-10 % | 5.50 (0.1397) | 7.00 (0.1778) | 14.00 (0.3556) | L7 | 100.6 | 2.6200 | 5.3992 | 0.00750 |

Differential gap is edge to edge. Uncontrolled default routing: 4.00 mil line and space (46.1 stripline / 64.8 microstrip ohm modelled), not impedance controlled.

## 7. Fabrication rules and capability

| Rule | Value |
|---|---|
| Minimum copper width | 3.00 mil |
| Minimum copper to copper space | 3.00 mil |
| Minimum copper to hole space | 6.00 mil |
| Minimum copper to board edge | 10.00 mil |
| Minimum annular ring | 2.50 mil |
| Minimum laser drill diameter | 4.00 mil |
| Minimum pitch - leaded package | 16.00 mil |
| Minimum pitch - BGA | 14.00 mil |
| Maximum board width | 18.0 in |
| Maximum board height | 24.0 in |
| Minimum silkscreen line width | 5.00 mil |
| Minimum silkscreen to soldermask space | 4.00 mil |
| Minimum silkscreen text height | 30.00 mil |
| Soldermask registration | 2.00 mil |
| Minimum soldermask opening | 3.00 mil |
| Minimum soldermask bridge | 4.00 mil |
| Minimum through-hole pad expansion outer | 4.00 mil |
| Minimum hole to hole | 10.00 mil |
| Minimum PTH pin to soldermask clearance | 8.00 mil |
| Minimum mechanical drill diameter | 6.00 mil |
| Maximum mechanical drill aspect ratio | 10.0:1 |
| Maximum laser microvia aspect ratio | 1.25:1 |
| Maximum stacked microvia levels | 2 |
| Minimum dielectric between adjacent copper layers | 2.50 mil |

## 8. Notes

This is a synthetic example document. Tamsworth Interconnect Co. is a fictional fabricator, and Corvel Coatings, Ashgrove Laminates and Brightwell Foil, with every product named here, are fictional too. The Dk, Df and roughness figures are illustrative and describe no real material.

Copper roughness is Rz in micrometres, matte and drum side.